

Meeting Notes — Prof. Asaf Cohen

Decentralized Detection: Questions & Insights -1

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1. The paper defines two cost functions: J^N for finite N and $J_{EE}^N = \frac{\log J^N}{N}$ for the infinite-sensor regime. What is the conceptual difference between them, and why does taking the logarithm give a meaningful objective?
2. Assumption 1(ii) requires that all observation distributions are dominated by a common base measure Q . What is the intuition behind this condition, and what concretely breaks in the analysis if it does not hold?
3. Assumption 1(iii) fixes a common action space $\mathbb{U} = \{1, \dots, M\}$ for all sensors, and Prof. Cohen's annotation suggests that allocating rate unevenly across sensors can improve performance. How exactly is that rate-allocation idea related to Assumption 1(iii), and is there a connection between the likelihood ratio l^i at each sensor and the total system rate R_{total} ?
4. Each sensor maps its observation to a discrete action u^i and sends it to the fusion center. Does the model assume sensors transmit immediately upon receiving a signal, or do they accumulate observations and send a packet?
5. Lemma 2 characterizes the optimal decision rule at the fusion center. What is the main point of the lemma, and how is it proven?
6. With M actions, the optimal sensor policy γ^i partitions the likelihood-ratio axis into M intervals. How does increasing the rate $R = \log_2 M$ affect the optimal policies γ^i for sensors $i > 1$?
7. The paper assumes sensors know the exact distributions $P_{Y|H_1}$ and $P_{Y|H_2}$ when designing their encoding policies. Can a *universal* sensor policy be designed that achieves optimal performance without knowledge of the true distributions — analogous to universal source coding?
8. The proof relies on observations being i.i.d. across sensors given the hypothesis. What happens to the exchangeability and identical-policy results when sensors are spatially correlated?
9. The paper considers binary hypothesis testing throughout. Do the main results — threshold policies and identical encoding in the limit — generalize to M -ary hypothesis testing?